

3	(a)	Despite both being at 100 °C , <u>steam transfers more (heat or thermal) energy</u> to the skin than boiling water. This is because <u>steam releases (heat) energy</u> proportional to the <u>latent heat of vaporisation</u> during condensation in addition to the heat energy released on cooling of condensed water on skin. This is much higher than the (heat) energy released by boiling water on contact with the skin.	B1 B1
	(b)(i)	Use state A: $n = \frac{P_A V_A}{RT_A} = \frac{20 \times 10^5 \times 1.2 \times 10^{-4}}{8.31 \times 800}$ $= 0.0361 \text{ mol}$ If state D is used, $n = 0.0362 \text{ mol}$.	C1 A1
	(b)(ii)1.	$T_B = \frac{P_B V_B}{R_B n} = \frac{20 \times 10^5 \times 5.2 \times 10^{-4}}{8.31 \times 0.0361}$ $= 3467 \text{ K}$ If $n = 0.0362 \text{ mol}$ is used, $T_B = 3457 \text{ K}$. OR $\frac{V_A}{T_A} = \frac{V_B}{T_B}$ $\frac{1.28}{800} = \frac{5.2}{T_B}$ $T_B = 3467 \text{ K}$	A1
	(b)(ii)2.	$\Delta U = \frac{3}{2} n R \Delta T$ $= \frac{3}{2} \times 0.0361 \times 8.31 \times (3467 - 800)$ $= 1200 \text{ J (allow ecf)}$ Accept $\frac{3}{2} p \Delta V = 1200 \text{ J}$.	C1 A1
	(b)(ii)3.	$W = -P \Delta V = -20 \times 10^5 \times (5.20 - 1.20) \times 10^{-4}$ $= -800 \text{ J}$ Award 1 mark for +800 J.	C1 A1
	(b)(ii)4.	From the first law of thermodynamics, $\Delta U = Q + W$ $Q = \Delta U - W$ $Q = 1200 - (-800) = 2000 \text{ J (allow ecf)}$	A1
	(b)(iii)	From D to A, since Q is zero, $W_{DA} = \Delta U_{DA}$ $W_{DA} = \frac{3}{2} \times 0.0361 \times 8.31 \times (800 - 226) = 258 \text{ J or } \underline{259 \text{ J}}$ if used $n = 0.0362$ $W_{\text{net by gas}} = 800 + 390 + 0 - 259 = 932 \text{ J or } \underline{931 \text{ J}}$ if used $n = 0.0362$ Allow ecf. Students can also use $W_{DA} = \frac{3}{2} (P_A V_A - P_B V_B) = 258 \text{ J}$	C1 C1 A1

4	(a)	In a polarised wave the vibrations of wave particles are limited to only one	B1
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